

Sequential Gaussian Avatars with Hierarchical Motion Context

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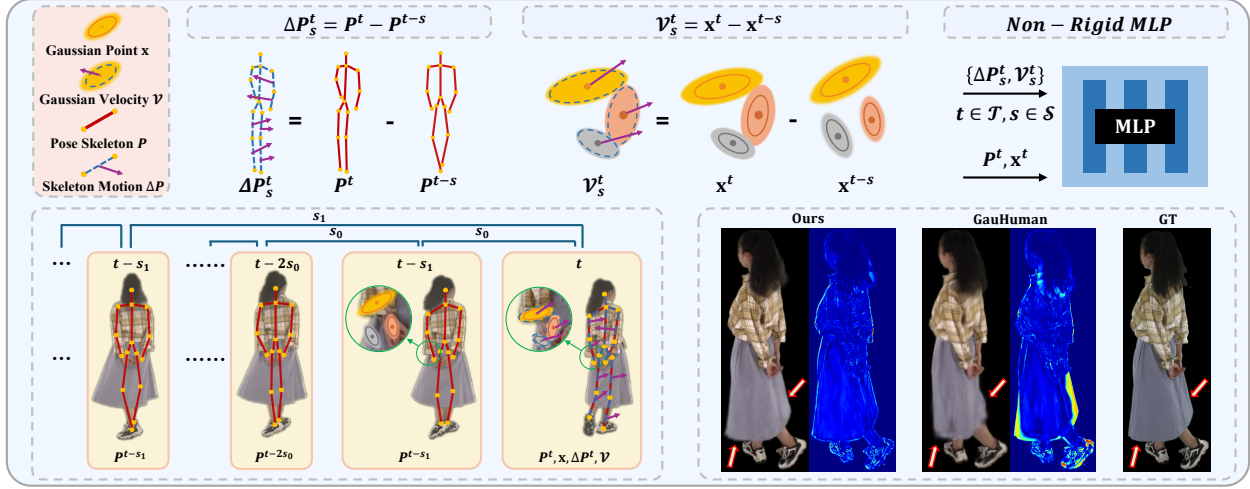


Figure 1. **Illustration of the Hierarchical Motion Context Design.** We model the motion-dependent appearance variations with both coarse skeleton condition ΔP and fine-grained point-wise velocity condition $\mathcal{V}$. ΔP describes the overall human skeleton motion, which is derived from the difference between the human poses at adjacent frames. $\mathcal{V}$ is the point-wise velocity that indicates finer-grained motion in local regions. To achieve robust deformation, we propose Spatio-temporal Multi-scale Sampling, which samples the overall motion trend and inter-frame details via diverse time intervals s for ΔP and $\mathcal{V}$.

Abstract

The emergence of neural rendering has significantly advanced the rendering quality of 3D human avatars, with the recently popular 3DGS technique enabling real-time performance. However, SMPL-driven 3DGS human avatars still struggle to capture fine appearance details due to the complex mapping from pose to appearance during fitting. In this paper, we propose SeqAvatar, which excavates the explicit 3DGS representation to better model human avatars based on a hierarchical motion context. Specifically, we utilize a coarse-to-fine motion conditions that incorporate both the overall human skeleton and fine-grained vertex motions for non-rigid deformation. To enhance the robustness of the proposed motion conditions, we adopt a spatio-temporal multi-scale sampling strategy to hierarchically integrate more motion clues to model human avatars. Extensive experiments demonstrate that our method significantly outperforms 3DGS-based approaches and renders human

avatars orders of magnitude faster than the latest NeRF-based models that incorporate temporal context, all while delivering performance that is at least comparable or even superior. Project page: <https://zezeaaa.github.io/projects/SeqAvatar/>

1. Introduction

Recent research on digital humans has highlighted the efficiency of 3D Gaussian Splatting (3DGS) [27], demonstrating its capability for high-quality and real-time rendering. By defining T-pose human Gaussians in a canonical space and employing specific warping techniques, we can render human avatars from diverse perspectives and in any pose. The canonical Gaussian primitives and warping weights are then jointly optimized under supervision from video inputs.

Animatable human avatar reconstruction is primarily hindered by limited modeling of non-rigid warping, forcing a trade-off between rendering quality and animation capability. Current methods [22, 56] incorporate SMPL(-X) pose priors [43, 51] to guide motion for each observation

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and rely on Linear Blend Skinning (LBS) for warping. Although these approaches are effective for pose-driven animation, they often struggle with per-frame non-rigid warping in scenarios involving complex garments.

We experimentally find that current human pose conditions [22, 56] do not fully capture the complex one-to-many mapping from pose to appearance, particularly for 3DGS-based approaches [27]. These methods [56, 68] rely on the spatial information of a template body in each frame’s human pose to predict non-rigid deformations. However, they often overlook local details, such as garment deformations far from the skeleton, and cannot resolve cases where the same pose corresponds to different appearances during complex motions. Furthermore, although previous NeRF-based [44] method [8] has attempted to model motion sequences using human pose residuals, the inherently global nature of the pose sequence limits the ability to capture finer motion details. Naive pose sequence modeling does not fully account for the explicit characteristics of 3DGS.

In this paper, we introduce a hierarchical motion context condition for 3DGS-based human avatar modeling to address the limitations of relying solely on human pose, which provides only limited global skeletal information. Our approach improves the ability of 3DGS-based methods to accurately capture the complex relationship between human pose and appearance in challenging scenarios. Specifically, we design a coarse-to-fine motion condition that incorporates both overall skeletal movements and fine-grained point-wise motions. Leveraging the explicit nature of Gaussian primitives, this condition seamlessly integrates into 3DGS-based methods, enabling more precise predictions of complex non-rigid deformations. To further enhance robustness, we propose a spatio-temporal multi-scale sampling strategy for constructing the motion context with a larger receptive field. Spatially, we consider the motion states of neighboring points within the local region of each Gaussian primitive to obtain more stable motion embeddings. Temporally, we capture human motion patterns across multiple time scales, combining long-term trends with fine-grained inter-frame motion details. This improves the model’s generalization to complex human movements.

To summarize, our key contributions are as follows:

- 1) We propose a novel hierarchical motion condition that integrates coarse-to-fine human motion, combining global skeletal poses with localized vertex residuals to enhance non-rigid deformation prediction.
- 2) We introduce a spatiotemporal multiscale sampling strategy that expands the receptive field of hierarchical motion context, improving generalization to complex motions.
- 3) Experiments on the I3D-Human [8], DNA-Rendering [9], and ZJU-MoCap [53] datasets show the effectiveness of the proposed **SeqAvatar**, which is capable of modeling details of the human body in complex motions.

2. Related Work

Neural Rendering. Neural rendering techniques have brought significant progress to human reconstruction and rendering. In particular, Neural Radiance Fields (NeRF) [44] introduces an implicit scene representation that models color and density using multilayer perceptrons, delivering photorealistic rendering results. More recently, point-based 3D Gaussian Splatting [27] utilizes 3D Gaussians to represent scenes explicitly, achieving real-time high-quality rendering. Building upon NeRF and 3DGS, subsequent works have advanced neural rendering across various dimensions, *e.g.*, enhancing visual fidelity [1, 2], improving sparse-view reconstruction quality [10, 52, 63, 71, 80], enabling pose-free optimization [37, 58, 61, 67], modeling dynamic scenes [13, 34, 49, 55, 69, 74], and accelerating training and inference [4, 12, 18, 39, 46, 60]. These developments also benefit human avatar modeling, which demands high-quality and efficient rendering.

SMPL(-X)-Based Neural Human Modeling. The SMPL(-X) family [43, 51] provides a parametric representation of the human body by decomposing it into pose-related and shape-related components using 3D mesh scans and principal component analysis (PCA). Its *pose blend shapes* enables body deformation through joint-wise pose blending, offering an efficient and compact way for animation. SMPL(-X) has thus become a cornerstone in human body modeling and animation, with many methods [11, 59] estimating its parameters directly from 2D inputs. Recent neural human reconstruction methods integrate SMPL(-X) with implicit representations such as NeRF [44] and 3DGS [27] to enable animatable avatars with high rendering quality. Different from purely 2D image animation methods [48, 62, 65, 73], these methods typically register the input data to a canonical T-pose space and use linear blend skinning (LBS) to transform 3D points into observation space based on SMPL(-X) poses. NeRF-based approaches [5–8, 14, 17, 19, 20, 30, 53, 64, 68, 75] focus on monocular or multi-view reconstruction. More recently, 3DGS-based methods [21–23, 25, 26, 29, 31–33, 36, 40, 41, 45, 47, 56, 76, 78, 79, 81] have gained popularity due to their real-time rendering and high fidelity. Some works [8, 23] further incorporate pose sequences as temporal context, but they suffer from slow rendering and lack fine details of motion caused by simply integrating coarse human pose embedding with NeRF.

Human Rendering with Temporal Embeddings The earliest modeling of dynamic radiance field [3, 13, 15, 16, 24, 34, 35, 42, 50, 57, 70] can be traced back to temporal embeddings in general scenes, where each frame’s observation is obtained by constructing a canonical space and a time-conditioned deformation field. Motivated by these, a stream of research [38, 72] focuses on pure rendering quality, employing temporal embeddings instead of human pose to en-

code each frame of human videos. Although these methods achieve high-quality, temporally continuous rendering, they struggle with the lack of geometric constraints from human pose, making parameterized animation unfeasible.

3. Preliminary

SMPL(-X) Series [43, 51] adopt a parameterized framework to represent human bodies across diverse shapes and poses. For each frame, the human mesh is derived by deforming a canonical template mesh based on shape and pose parameters. Specifically, a 3D point $\mathbf{x}$ on the canonical mesh is warped to the corresponding point on the deformed mesh in the observation space, following

$$\text{LBS}(\mathbf{x}, \mathbf{B}_k, \omega_k(\mathbf{x})) = \sum_{k=1}^K \omega_k(\mathbf{x}) \mathbf{B}_k \mathbf{x}, \quad (1)$$

where K is the total bone number and $\mathbf{B}_k$ is the transformation matrix for each bone. Specifically, $\mathbf{B}_k$ consists of a rotation and translation matrix $\mathbf{R}_k$ and $\mathbf{b}_k$. Here, $\mathbf{R}_k$ represents the global rotation of each joint, influenced by the local joint rotations $\mathbf{r}$, while $\mathbf{b}_k$ denotes each joint’s translation, determined by the joint positions $\mathbf{j}$ and the human shape parameters β . The linear blending weight ω^k depends on $\mathbf{x}$ and is regressed from comprehensive human meshes.

3D Gaussian Splatting [27] utilizes a set of point-based Gaussian primitives to explicitly represent scenes, enabling real-time and high-quality rendering. Each Gaussian primitive is defined by its center position $\mathbf{x} \in \mathbb{R}^3$ and covariance matrix $\Sigma \in \mathbb{R}^{3 \times 3}$. To ensure positive semi-definiteness and simplify optimization, the covariance matrix Σ is decomposed into a rotation matrix $\mathbf{R}$ and a scaling matrix $\mathbf{S}$:

$$\Sigma = \mathbf{R} \mathbf{S} \mathbf{S}^T \mathbf{R}^T, \quad (2)$$

where $\mathbf{R}$ and $\mathbf{S}$ are derived by a scaling vector $\mathbf{s} \in \mathbb{R}^3$ and a quaternion vector $\mathbf{r} \in \mathbb{R}^4$ in practice. Additionally, each Gaussian primitive is also assigned a color feature $sh \in \mathbb{R}^k$ represented by spherical harmonics (SH) and an opacity α for rendering. During the rendering process, a splatting technique [27] is used to project each Gaussian primitive onto the 2D image space with a viewing transform $\mathbf{W}$ and the Jacobian $\mathbf{J}$ of the projective transformation’s affine approximation. The transformed covariance Σ' in the camera’s coordinates is

$$\Sigma' = \mathbf{J} \mathbf{W} \Sigma \mathbf{W}^T \mathbf{J}^T. \quad (3)$$

After projection, the pixel color C is computed by blending N ordered Gaussian primitives overlapping at the pixel:

$$C = \sum_{i \in N} c_i \alpha'_i \prod_{j=1}^{i-1} (1 - \alpha'_j), \quad (4)$$

where c_i is computed from SH feature sh and α'_i is the product of α_i and probability density of i -th 2D Gaussian.

For optimization, a photometric loss is defined by a combination of L_1 and SSIM [66] losses:

$$L_{photo} = \lambda L_1(\hat{I}, I) + (1 - \lambda)(1 - SSIM(\hat{I}, I)), \quad (5)$$

where $\hat{I}$ and I denote the rendered and ground-truth images, and λ controls the balance between the two terms.

4. Method

Fig. 2 shows the framework of our method. We use the explicit point-based 3DGS as the representation of the human body. Given a collection of input cameras and images, we optimize a set of Gaussian primitives $\{\mathcal{G}_i\}_{i=1}^{i=n}$ to fit the body’s shape and appearance. Each Gaussian primitive $\mathcal{G}_i$ includes the center position $\mathbf{x}$, scaling vector $\mathbf{s}$, rotation quaternion vector $\mathbf{r}$, color feature sh and opacity α , where $\mathbf{x}$ is initialized from the SMPL template vertices $\{\mathbf{T}_i\}_{i=1}^{i=N}$. During the optimization process, we first apply a coarse-to-fine motion context to capture more accurate and fine-grained details (Sec. 4.1). To mitigate overfitting, we propose Spatio-Temporal Multi-Scale Sampling to obtain more robust point-wise Gaussian motion, which serves as the embedding for non-rigid deformation (Sec.4.2). Next, we adopt Linear Blend Skinning (LBS) to map canonical Gaussian primitives $\mathcal{G}$ to observation space and render images via differentiable splitting (Sec.4.3).

4.1. Non-Rigid Deformation with Coarse-to-fine Motion Context

Coarse Skeleton Motion. As illustrated in Fig. 1, we introduce a coarse-to-fine motion context to excavate additional conditions for non-rigid deformation. Typically, most human 3D avatar methods use the current frame pose as a condition to predict non-rigid deformation [22, 56]. While this provides spatial information to distinguish the motion of different body parts, it fails to capture temporal motion changes, as discussed in [8]. Therefore, given a frame at time t , we consider a sequence of regularly interval-sampled frames $\mathcal{T}$ to model inter-frame body motion variations:

$$\mathcal{T} = \{t - s, t - 2s, \dots, t - Ls\}, \quad (6)$$

where L is the sequence length and s is the time interval. The coarse motion of body skeletons at each time step $t \in \mathcal{T}$ can be derived by calculating the difference of poses between adjacent frames as illustrated in Fig. 1, following

$$\Delta \mathcal{P} = \{\Delta P^t = \delta(P^t, P^{t-s}) | t \in \mathcal{T}\}, \quad (7)$$

where $P \in \mathbb{R}^{K \times 3}$ is the body pose and δ denotes the difference $\Delta \mathcal{P}$ between two poses in axis-angle form. We employ an MLP $\mathcal{E}_{\Delta \mathcal{P}}$ to encode the sequential skeleton mo-

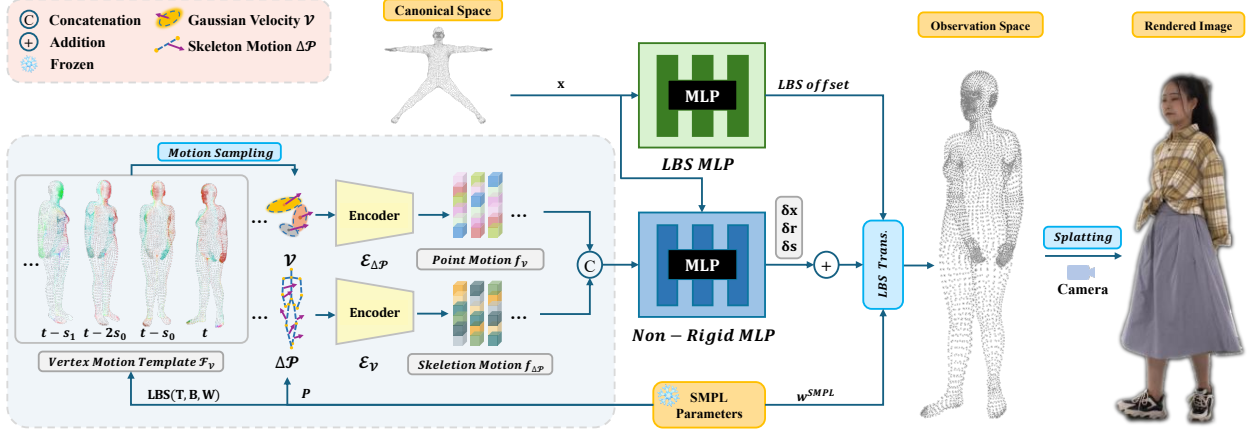


Figure 2. **Overview of the proposed method.** We first initialize canonical Gaussian positions $\mathbf{x}$ with SMPL template vertices. For each Gaussian, we derive both coarse skeleton motion condition $f_{\Delta\mathcal{P}}$ and fine-grained vertex motion condition $f_{\mathcal{V}}$ that sampled from the vertex motion template $\mathcal{F}_{\mathcal{V}}$ (points in different colors represent different motions). Based on such hierarchical motion information, we utilize an MLP $\mathcal{E}_{non-rigid}$ to better predict each Gaussian’s non-rigid deformation prediction. The non-rigid deformed Gaussians are then warped into observation space via the standard LBS transformation for rendering.

tion $\Delta\mathcal{P} \in \mathbb{R}^{L \times K \times 3}$, flattened along the temporal dimension, into a skeleton motion embedding $f_{\Delta\mathcal{P}} \in \mathbb{R}^{32}$,

$$f_{\Delta\mathcal{P}} = \mathcal{E}_{\Delta\mathcal{P}}(\Delta\mathcal{P}), \quad (8)$$

which serves as a condition for the subsequent non-rigid deformation. Further details are provided in the supplementary material.

Fine Vertex Motion. Compared to previous implicit NeRF-based methods, the point-based 3DGS representation enables us to explore more fine-grained temporal motion information. We derive a point-wise velocity vector v_i for each Gaussian primitive $\mathcal{G}_i$ to model the fine-grained localized body motion, which is beyond the scope of pose skeleton motion $\Delta\mathcal{P}$. By measuring the position variations across adjacent time steps, we calculate per-frame velocity

$$v^t = \frac{\mathbf{x}_o^t - \mathbf{x}_o^{t-s}}{s}, \quad (9)$$

where $\mathbf{x}_o$ denotes the warped coordinates in observation space. However, since the positions of Gaussian primitives are updated during optimization, it is not stable to obtain v_i based on such a dynamic variable. Moreover, transforming the canonical Gaussians into the observation space requires non-rigid deformation first, which leads to a circular dependency and conflicts with the current velocity design. To this end, we introduce a motion template field $\mathcal{F}_{\mathcal{V}} = \{\mathbf{V}_i\}_{i=1}^N$ that stores the velocity of each SMPL vertex, and v_i for each Gaussian primitive is sampled from $\mathcal{F}_{\mathcal{V}}$ based on the distance between query points and vertices in this template. Specifically, given a time step t and the corresponding body pose $P^t = \{(\mathbf{R}_k^t, \mathbf{b}_k^t) | k \in K\}$, the SMPL template vertices in $\mathbf{T}$ can be warped into observation space with the template skinning weights $\mathbf{W}$ and

standard linear blend skinning, following

$$\mathbf{T}_o^t = \text{LBS}(\mathbf{T}, \mathbf{B}^t, \mathbf{W}), \quad (10)$$

where $\mathbf{B}^t$ is the transformation matrix derived from rotation and translation matrix $\mathbf{R}^t$ and $\mathbf{b}^t$. $\mathbf{W}$ is the template SMPL LBS weights. The velocity $\mathbf{V}$ of each template vertices $\mathbf{T}$ can be further derived as

$$\mathbf{V}^t = \frac{\mathbf{T}_o^t - \mathbf{T}_o^{t-s}}{s}. \quad (11)$$

We then sample each Gaussian’s velocity from this motion template field $\mathcal{F}_{\mathcal{V}}$, which is discussed in Sec. 4.2.

4.2. Spatio-temporal Multi-scale Sampling (STMS)

To enhance the robustness of human motion conditions, we propose modeling both local region motion information of the human body and motion patterns across different temporal windows to mitigate overfitting and improve generalization by capturing more comprehensive motion dynamics.

In the spatial dimension, we sample the τ nearest template to model the body’s local region motion more robustly. Specifically, for each canonical Gaussian primitive $\mathcal{G}_i$, the τ nearest vertices’ velocities are sampled as input to an MLP $\mathcal{E}_{knn}$ to learn a local point-wise motion embedding

$$e_i^t = \mathcal{E}_{knn}(\{\mathbf{V}_j^t\}), \quad j \in \text{KNN}(\mathbf{T}, \mathbf{x}_i), \quad (12)$$

where $\text{KNN}(\mathbf{T}, \mathbf{x}_i)$ denotes the τ nearest SMPL template vertices of the canonical Gaussian position $\mathbf{x}_i$. Similar to Eq. 7, we then apply an MLP $\mathcal{E}_{\mathcal{V}}$ to encode each Gaussian primitive’s sequential motion embedding $\mathcal{V} = \{e^t | t \in \mathcal{T}\}$ into a point-wise sequential condition $f_{\mathcal{V}} \in \mathbb{R}^{96}$

$$f_{\mathcal{V}} = \mathcal{E}_{\mathcal{V}}(\mathcal{V}). \quad (13)$$

In the temporal dimension, to capture both the overall motion trend and the inter-frame motion details, we adopt a multi-scale sequence sampling strategy. In detail, we sample a series of sequences at several progressively increasing intervals to get human motion information across different temporal windows

$$\mathcal{S} = \{s = s_0 + i\Delta s\}_{i=0}^{i=m}, \quad (14)$$

where Δs is the increasing rate of sampling interval s . We then input the multi-scale sampled sequence motions into the skeleton motion encoder $\mathcal{E}_{\Delta\mathcal{P}}$ and the point-wise motion encoder $\mathcal{E}_{\mathcal{V}}$ to obtain hierarchical temporal motion embeddings. Therefore, Eq. 7 and Eq. 13 can be revised as

$$f_{\Delta\mathcal{P}} = \mathcal{E}_{\Delta\mathcal{P}}(\{\Delta\mathcal{P}_s\}), \quad s \in \mathcal{S} \quad (15)$$

$$f_{\mathcal{V}} = \mathcal{E}_{\mathcal{V}}(\{\mathcal{V}_s\}), \quad s \in \mathcal{S}. \quad (16)$$

In practice, we concatenate all skeleton motion conditions, $\Delta\mathcal{P}_s$ for $s \in \mathcal{S}$, and localized vertex motion conditions, $\mathcal{V}_s$ for $s \in \mathcal{S}$, across different sampling scales. These are then input to $\mathcal{E}_{\Delta\mathcal{P}}$ and $\mathcal{E}_{\mathcal{V}}$, respectively. More details are shown in the supplementary material.

Non-Rigid Deformation. Given the coarse skeleton motion $f_{\Delta\mathcal{P}} = \mathcal{E}_{\Delta\mathcal{P}}(\{\Delta\mathcal{P}_s\})$ condition and fine point-wise velocity condition $f_{\mathcal{V}} = \mathcal{E}_{\mathcal{V}}(\{\mathcal{V}_s\})$, we utilize an MLP to predict each Gaussian’s non-rigid deformation as

$$\delta\mathbf{x}, \delta\mathbf{s}, \delta\mathbf{r} = \mathcal{E}_{non-rigid}(\mathbf{x}, P, f_{\Delta\mathcal{P}}, f_{\mathcal{V}}). \quad (17)$$

The deformed canonical Gaussian $\mathcal{G}'$ is

$$\mathbf{x}' = \mathbf{x} + \delta\mathbf{x}, \quad (18)$$

$$\mathbf{s}' = \mathbf{s} + \delta\mathbf{s}, \quad (19)$$

$$\mathbf{r}' = \mathbf{r} \cdot \delta\mathbf{r}, \quad (20)$$

where $\cdot$ denotes the multiplication of two quaternions.

4.3. Optimization

Rigid Deformation. We utilize the standard LBS operation to map the non-rigid deformed Gaussians $\mathcal{G}_o$ into the observation space, following

$$\mathbf{x}_o = \mathbf{LBS}(\mathbf{x}', \mathbf{B}, \omega), \quad (21)$$

$$\mathbf{R}_o = \sum_{k=1}^K \omega_k(\mathbf{x}') \mathbf{B}_k \mathbf{R}', \quad (22)$$

where $\mathbf{R}'$ is the rotation matrix derived from the non-rigid deformed Gaussian’s rotation quaternion $\mathbf{r}'$, and $\mathbf{R}_o$ represents the rotation matrix in observation space. Following the previous method [22], we utilize an MLP $\mathcal{E}_{lbs}$ to predict the LBS weight offsets for each query canonical point and update the sampled weights from the nearest SMPL vertex as

$$\omega_k(\mathbf{x}) = \omega_k^{SMPL}(\mathbf{x}) + \mathcal{E}_{lbs}(\mathbf{x}). \quad (23)$$

Similar to [8, 22, 68], we introduce a pose refinement MLP $\mathcal{E}_{pose}$ to refine the pose estimate from SMPL for a better fit to the human body.

Loss Function. With the transformed Gaussian primitives $\mathcal{G}_o$ in observation space, we apply the standard splitting [27] to render images as

$$I = \text{Splatting}(\mathbf{x}_o, \mathbf{R}_o, \mathbf{s}', \alpha, sh). \quad (24)$$

During the optimization process, we employ a combination of loss functions as supervision, summarized as

$$\mathcal{L} = \lambda_1 \mathcal{L}_{color} + \lambda_2 \mathcal{L}_{ssim} + \lambda_3 \mathcal{L}_{lips} + \mathcal{L}_{mask}, \quad (25)$$

where λ is used to balance the weight of different losses. $\mathcal{L}_{mask}$ [22] is an L_2 loss between the rendered α and human body mask. Similar to [56], we apply $\mathcal{L}_{isopos}$ and $\mathcal{L}_{isocov}$ to control the Gaussian primitive’s position and covariance. Details are shown in the supplementary material.

5. Experiments

5.1. Datasets

DNA-Rendering [9] is a challenging human-centric rendering dataset. It contains diverse scenes from everyday life to professional occasions. We use 6 sequences with loose-fitting garments (1_0206_04, 2_0007_04, 2_0019_10, 2_0044_11, 2_0051_09, 2_0813_05) for experiments. We adopt 24 views for training and 6 views for testing.

I3D-Human [8] contains scenes closer to daily life, which comprises multi-view video frames of humans with loose clothing and complex movements. We conduct experiments on 4 sequences and use 4-5 views for training and the rest for testing. All training and testing data are retained to be the same in comparisons following Dyco’s [8] data split.

5.2. Baselines and Metrics

We compare against the state-of-the-art human modeling methods including Dyco [8], 3DGS-Avatar [56], GART [31], and GauHuman [22]. Dyco is an implicit NeRF-based method that adopts pose sequence information for temporal modeling. We compare performance with Dyco to showcase the better efficacy of our fine-grained motion condition design. 3DGS-Avatar, GART, and GauHuman are explicit 3DGS-based methods, and all of them are originally designed for monocular inputs. For fairness, we extend them to multi-view input under the same settings. We report three key metrics: peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM) [66], and learned perceptual image patch similarity (LPIPS) [77].

5.3. Comparison

Comparison on DNA-Rendering. Tab. 1 shows the quantitative results on DNA-Rendering dataset. The proposed

Table 1. **Quantitative Results on DNA-Rendering Dataset.** We report the performance of novel view rendering. Our method outperforms previous state-of-the-art human modeling methods. We mark the **best** and the **second best** methods in cells. $LPIPS^* = LPIPS \times 10^3$.

Subject:	1_0206_04			2_0007_04			2_0019_10			2_0044_11			2_0051_09			2_0813_05		
Metric:	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $^*\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $^*\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $^*\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $^*\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $^*\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $^*\downarrow$
3DGS-Avatar [56]	26.64	0.9439	50.09	28.06	0.9492	56.70	32.06	0.9710	31.12	28.23	0.9512	37.15	25.04	0.9534	42.22	31.75	0.9701	31.30
GART [31]	27.19	0.9454	56.47	28.22	0.9525	56.46	31.13	0.9686	35.16	28.90	0.9561	41.27	26.47	0.9628	42.21	32.06	0.9729	35.74
GauHuman [22]	27.96	0.9500	50.42	27.93	0.9496	56.93	31.63	0.9698	32.82	30.38	0.9641	34.19	25.99	0.9599	42.00	33.43	0.9764	29.42
Ours	30.81	0.9649	39.47	29.63	0.9566	43.72	34.97	0.9777	23.53	32.62	0.9747	23.83	28.45	0.9685	34.06	35.81	0.9843	20.83

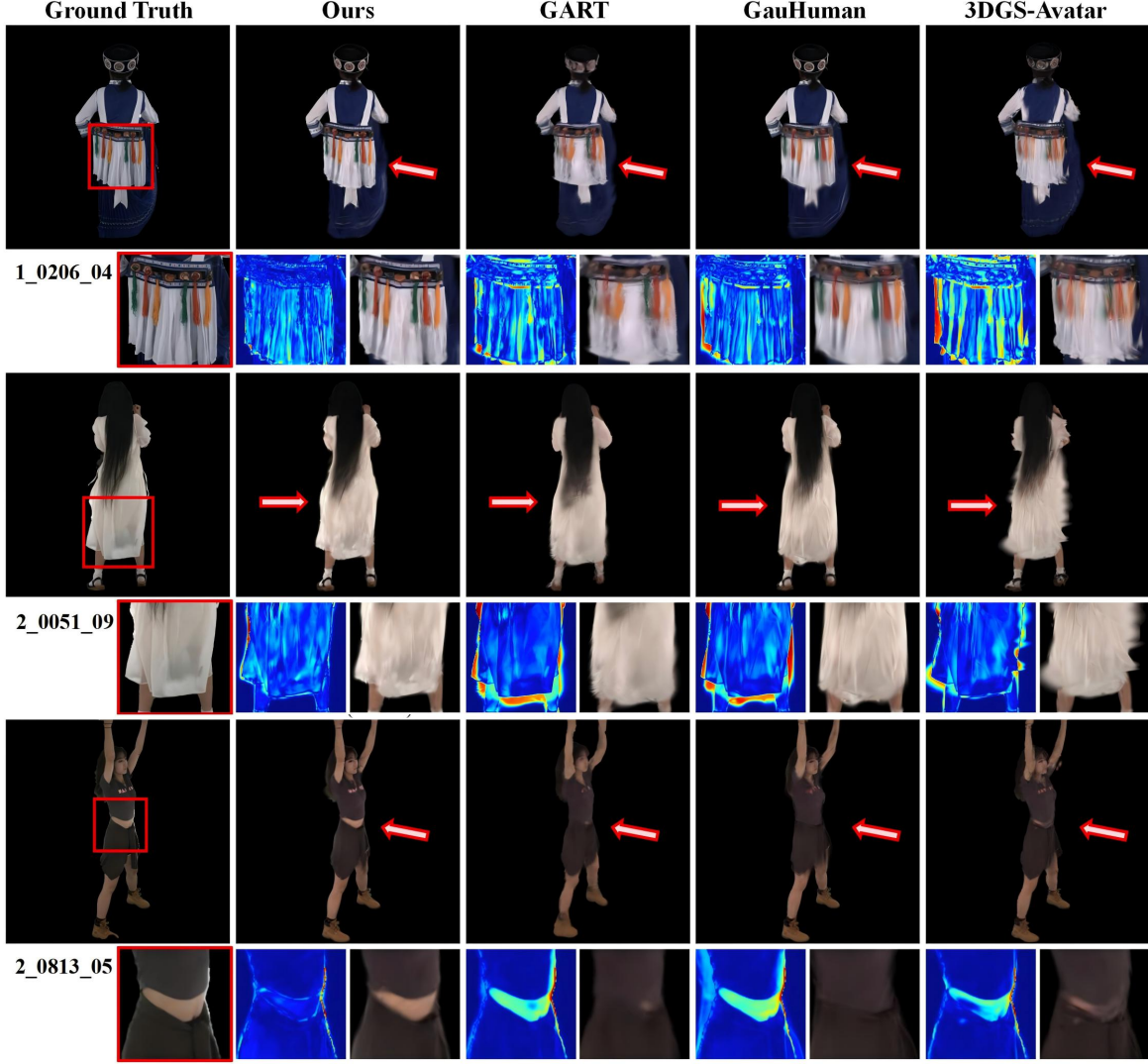


Figure 3. **Novel View Qualitative Results on DNA-Rendering.** We zoom into the local region and compute the error maps compared with ground truth images. The results show that our method achieves competitive results on both the overall and local region qualities.

method outperforms previous state-of-the-art human modeling methods across all metrics. To demonstrate the visual improvement, we compare the quality of rendering images in Fig. 3. Previous methods, which lack the temporal motion information for non-rigid deformations, are unable to predict the movement of Gaussian primitives properly, resulting in blurred renderings in regions with complex tex-

tures (scene 1_0206_04 in Fig. 3). Besides, the results in Fig. 3 show that the compared methods fail to capture the proper shape variation caused by movement (e.g., the white dress in scene 2_0051_09). Although these methods achieve a smooth appearance in these areas, the error maps show that they fail to match the actual shape closely. We leverage the hierarchical motion context to predict Gaussian defor-

Table 2. Novel View Rendering Quantitative Results on I3D-Human.

Subject:	ID1_1			ID1_2			ID2_1			ID3_1		
Metric:	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
3DGS-Avatar [56]	29.78	0.9600	31.40	31.29	0.9627	28.09	29.63	0.9604	37.33	32.73	0.9602	39.47
Dyco [8]	30.81	0.9617	28.29	31.45	0.9628	25.75	29.41	0.9618	33.26	32.50	0.9612	35.46
GauHuman [22]	29.47	0.9571	39.46	30.42	0.9585	38.87	28.51	0.9546	49.46	32.10	0.9546	53.67
Ours	31.81	0.9659	27.03	31.98	0.9660	27.84	31.53	0.9685	30.23	33.62	0.9651	34.02

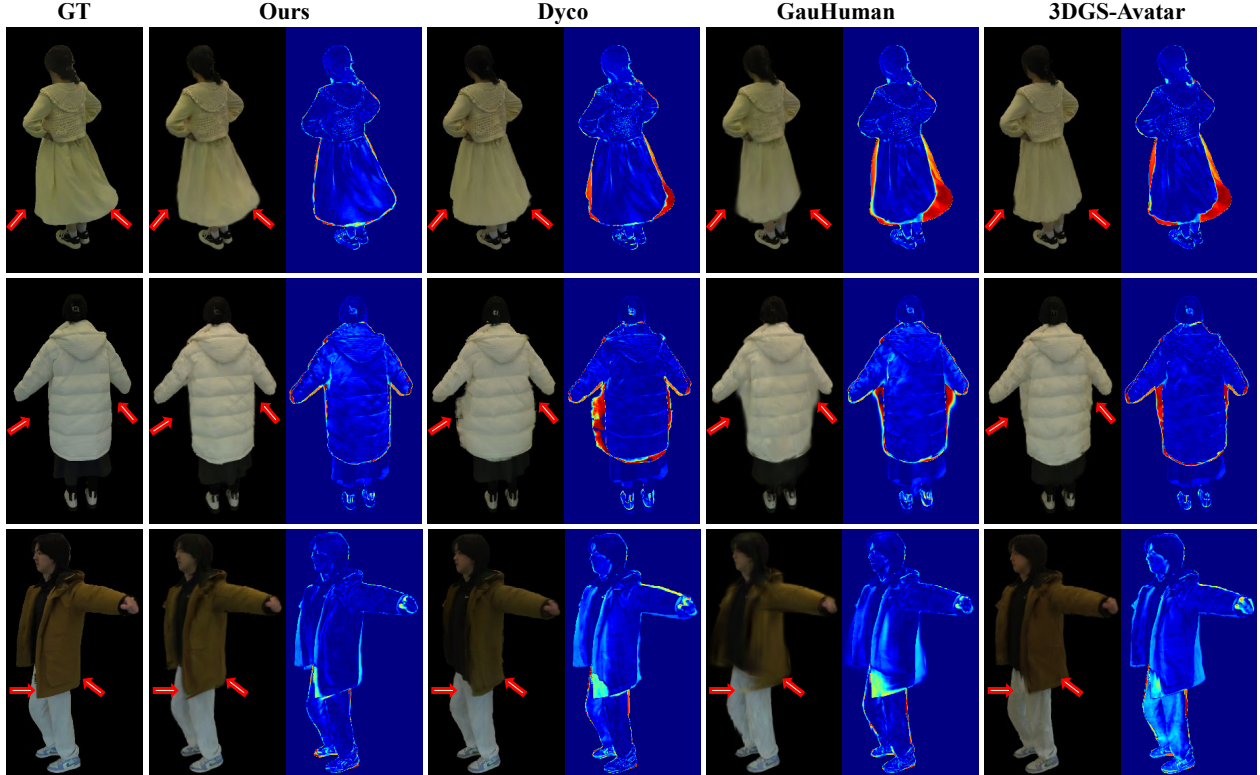


Figure 4. **Novel Pose Qualitative Results on I3D-Human.** We compare the proposed method with previous SOTA approaches. The rendered images and error maps demonstrate our robustness on novel pose rendering.

mations, allowing our method to capture detailed appearance variations caused by human motions more accurately.

Comparison on I3D-Human. The quantitative results in Tab. 2 show that our method surpasses the previous SOTA method on most metrics, demonstrating our competitive performance on the I3D-Human dataset. Previous NeRF-based Dyco [8] utilizes pose variation as a condition to model human motions. However, it is limited to capturing only overall body motions and lacks the capacity for finer-grained modeling in local regions. For example, in ID1.1 and ID2.1 shown in Fig. 8 (in the Supp. Mat.), our method renders results that accurately reflect the true motion, while Dyco fails to model the information in regions far from the human body skeleton. Tab. 3 reports the quantitative results of novel pose rendering on the I3D-Human dataset. Our method achieves best performance among 3DGS-based human modeling methods and also enables real-time render-

ing (~ 45 FPS on I3D-Human) compared to NeRF-based Dyco (~ 0.7 FPS). Fig. 4 shows the qualitative comparisons of novel pose rendering. Different from Dyco [8], which relies solely on joint motions as conditions, our method leverages the motion state of each Gaussian primitive to predict deformations, enabling more flexible modeling of motions in regions distant from the human body.

To evaluate the generalizability, we also conduct experiments on ZJU-MoCap [53]. Please refer to the supplementary materials for more details.

Out-of-Distribution Poses. To evaluate the generalization ability to novel poses, we conduct experiments using animations with large pose variations. Specifically, we use a model trained on one sequence to render target poses sampled from other unseen sequences, as different sequences in DNA-Rendering and I3D-Human contain distinct motion patterns. As illustrated in Fig. 6, the *Target Pose* is

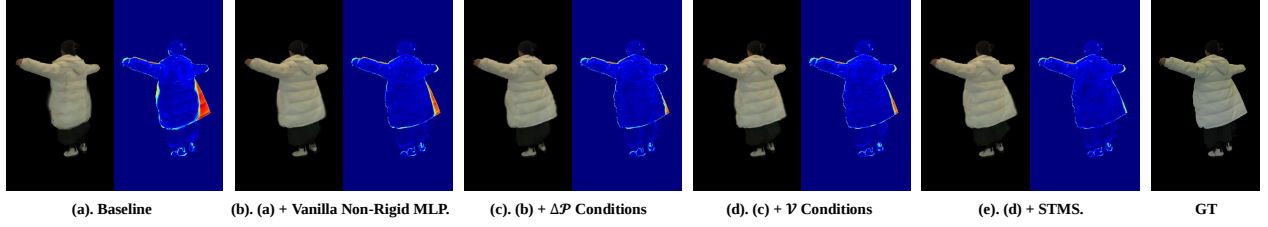


Figure 5. **Qualitative Ablation Results.** We compare the visual influence of different components on ID2_1 scene of I3D-Human dataset.

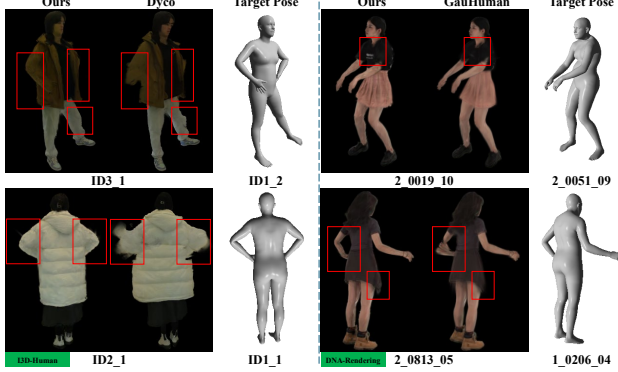


Figure 6. **Out-of-distribution Pose Animation.**

drawn from an unseen sequence exhibiting different human motions. The results suggest that our method is capable of generalizing to these significant pose changes, even though such poses were not observed during training.

Table 3. **Metrics of Novel Pose Renderings on I3D-Human.**

Methods	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS* $\downarrow$
Dyco [8]	30.09	0.9569	35.32
3DGS-Avatar [56]	29.54	0.9555	38.59
GauHuman [22]	29.31	0.9526	48.29
Ours	30.28	0.9583	36.07

5.4. Ablation Study

In this section, we ablate the influence of various components in our methods on I3D-Human dataset. Tab. 4 shows the average metrics of 4 sequences under different settings. **Skeleton motion condition** $\Delta\mathcal{P}$ denotes the overall human body movement. Compared to predicting non-rigid deformation only with the Gaussian position, incorporating this additional global temporal information allows our method to better model the overall human shape and achieve higher rendering quality, as demonstrated in Tab. 4 (c).

Fine-grained point-wise motion condition $\mathcal{V}$ describes the local region’s variation in a more detailed manner. Such point-based motion information captures the relationship between motion and appearance variations in areas beyond the human skeleton. Tab. 4 (d) and Fig. 5 (d) show that our

method achieves higher rendering details in local regions. **Spatio-temporal Multi-scale Sampling (STMS)** is designed to enhance the robustness of human motion conditions by incorporating local region motion information and motion patterns across different temporal windows. With such a hierarchical motion context embedding design, our method is able to predict each Gaussian’s deformation more robustly, leading to better rendering quality as shown in Tab. 4 (e) and Fig. 5 (e). Please refer to the supplementary materials for more detailed ablation experiments.

Table 4. **Quantitative Ablation Results on I3D-Human.** (a) denotes experiments without non-rigid deformation. (b) only utilizes Gaussian positions $\mathbf{x}$ and the current frame pose P as the conditions for non-rigid MLP.

Methods	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS* $\downarrow$
(a) Baseline	29.76	0.9569	38.35
(b) (a)+Vanilla Non-Rigid MLP+ $\mathcal{P}$ Conditions.	31.05	0.9617	34.35
(c) (b)+ $\Delta\mathcal{P}$ Conditions.	31.89	0.9645	32.17
(d) (c)+ $\mathcal{V}$ Conditions.	32.01	0.9651	31.23
(e) (d)+STMS. (Ours)	32.24	0.9664	29.78

6. Limitations and Conclusion

Limitations. The Gaussian-based representation used in our method may introduce slight blur artifacts during rendering, whereas NeRF’s ray-based volumetric integration tends to produce sharper results. Moreover, our local velocity cues are derived from the coarse SMPL model rather than dense surface tracking [79], which may limit the accuracy of fine-scale garment deformation. Addressing these limitations remains an open challenge for future work.

Conclusion. In this paper, we propose a 3DGS-based framework that integrates hierarchical motion context for 3D human modeling. Specifically, the non-rigid deformation for the Gaussian primitive is learned based on the global human skeleton variations and fine-grained Gaussian’s point-wise motions. To capture each Gaussian primitive’s motion information more robustly, we introduce a spatio-temporal multi-scale sampling strategy to incorporate both local region motion features and motion patterns across different temporal intervals. Through the above design, the proposed method achieves state-of-the-art rendering of human avatars with complex garments and motions.

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Sequential Gaussian Avatars with Hierarchical Motion Context

Supplementary Material

7. Details of Skeleton Motion ΔP

The pose P is represented as the rotation relative to the parent node of all joints:

$$P = \{\theta_1, \theta_2, \dots, \theta_K\}, \quad (26)$$

where $\theta_j \in \mathbb{R}^3$ describes the j -th joint's relative rotation with its parent in axis-angle form [43]. We follow Dyco [8] to derive each joint's rotation variation $\Delta\theta$ between adjacent frames as its skeleton motion representation. Specifically, the rotation of j -th joint in matrix form can be obtained according to *Rodrigues formula* [43]:

$$\mathbf{R}_j = \text{Rodrigues}(\theta_j), \quad (27)$$

where $\mathbf{R}_j$ is a 3×3 rotation matrix. Given 2 consecutive poses P^t and P^{t-s} , the temporal variation of j -th joint's rotation is computed as a relative transformation matrix $\Delta\mathbf{R}_j^t$:

$$\Delta\mathbf{R}_j^t = \mathbf{R}_j^t \mathbf{R}_j^{t-s-1}. \quad (28)$$

Then we convert the 3×3 matrix $\Delta\mathbf{R}_j^t$ back to the axis-angle form $\Delta\theta_j^t \in \mathbb{R}^3$. Now, the skeleton motion at time t is

$$\Delta P^t = \{\Delta\theta_1^t, \Delta\theta_2^t, \dots, \Delta\theta_K^t\}, \quad (29)$$

which is simplified as $\Delta P^t = \delta(P^t, P^{t-s})$ in Eq. (7).

8. Datasets

I3D-Human Dataset [8]. We use images at a resolution of 512×512 for experiments and follow Dyco[8] for both training and testing splits.

DNA-Rendering Dataset [9]. We use images with a resolution of 512×612 for training and 750×1024 for testing in our experiments. For the temporal split, we select 100 consecutive time steps (pose IDs 0–99) and use the multi-view images of all selected steps for training. During testing, we evaluate on a total of 20 time steps sampled every 5 steps from this training range, using multi-view images from held-out camera views. For the view split, we uniformly sample 24 views from camera IDs 0–48 for training and 6 additional views uniformly sampled from camera IDs 48–60 for novel view evaluation.

ZJU-MoCap [53]. We use images with resolution 512×512 for experiments and follow 3DGS-Avatar[56] for both training and testing splits.

9. Motion Embeddings for Non-Rigid MLP

The embeddings $f_{\Delta P}$ and f_V in Eq. (17) describe the skeleton and local region motions, respectively.

For each frame, all Gaussian primitives $\{\mathcal{G}_i\}$ share the same skeleton motion embedding $f_{\Delta P} \in \mathbb{R}^{32}$ and we concat it with $\mathcal{G}_i$'s point-wise motion embedding $f_{V_i} \in \mathbb{R}^{96}$ as a new condition $f'_i \in \mathbb{R}^{128}$. f'_i is then input to an MLP $\mathcal{E}_{non-rigid}$ to predict the corresponding non-rigid deformation. $\mathcal{E}_{non-rigid}$ consists of 3 hidden layers and one output layer, and each hidden layer is followed by a ReLU activation.

10. Details of Loss Functions

$\mathcal{L}_{color}$ is the L_1 loss between the rendered image I and the ground truth I_{gt} :

$$\mathcal{L}_{color} = |I - I_{gt}|. \quad (30)$$

We use $\mathcal{L}_{ssim}$ to constrain the structure similarity between the rendered image and the ground truth, which is given by

$$\mathcal{L}_{ssim} = 1 - \text{SSIM}(I, I_{gt}), \quad (31)$$

where $\text{SSIM}(\cdot)$ is the SSIM metric. Additionally, we use the LPIPS loss to ensure the perceptual similarity:

$$\mathcal{L}_{lrips} = \text{LPIPS}(I, I_{gt}), \quad (32)$$

where $\text{LPIPS}(\cdot)$ is the LPIPS metric. Following GauHuman [22] and 3DGS-Avatar [56], we employ a mask loss to ensure that Gaussian primitives are accurately localized within their designated regions:

$$\mathcal{L}_{mask} = \|M - M_{gt}\|_2 \quad (33)$$

where M_{gt} is the foreground mask and M is the accumulated α value:

$$M = \sum \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j). \quad (34)$$

Similar to [54, 56], we also utilize the as-isometric-as-possible constraint [28] to enforce neighboring distance similarity of 3D Gaussian centers and covariance matrices between canonical space and observation space:

$$\mathcal{L}_{isopos} = \sum_{i=1}^N \sum_{j \in \text{KNN}(i)} |d(\mathbf{x}^i, \mathbf{x}^j) - d(\mathbf{x}_o^i, \mathbf{x}_o^j)|, \quad (35)$$

$$\mathcal{L}_{isocov} = \sum_{i=1}^N \sum_{j \in \text{KNN}(i)} |d(\Sigma^i, \Sigma^j) - d(\Sigma_o^i, \Sigma_o^j)|, \quad (36)$$

where N is the number of Gaussian primitives. $\mathbf{x}, \Sigma$ and $\mathbf{x}_o, \Sigma_o$ are the Gaussian primitive's position and covariance matrix in canonical space and observation space, respectively. KNN denotes the $K = 5$ nearest points.

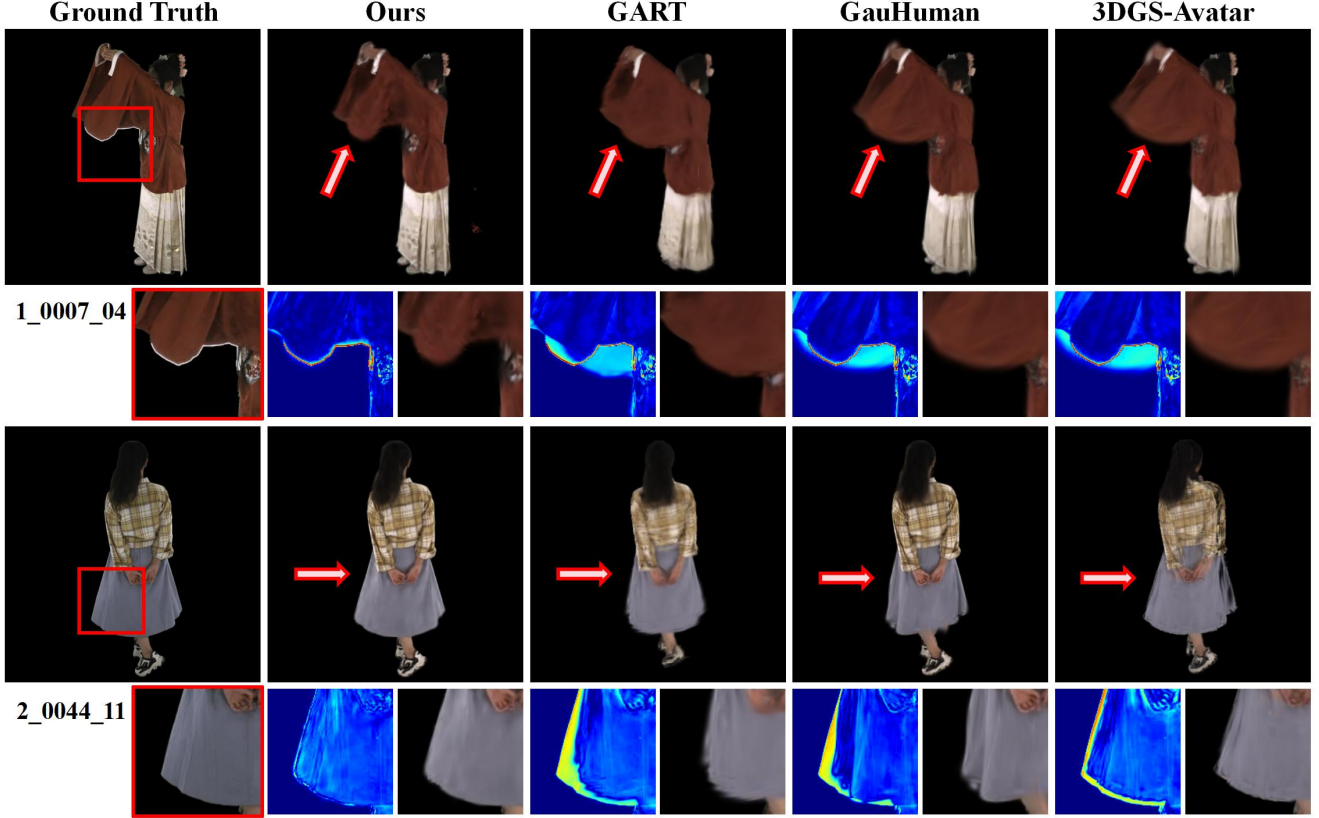


Figure 7. **Novel View Qualitative Results on DNA-Rendering.** We provide both the complete image and localized error map comparisons against other methods for a more comprehensive visualization.

11. Details of Optimization

In our experiments, the number $|\mathcal{S}|$ (in Eq. (14)) of sequence sampled in different scales is set to 3, and each sequence length L (in Eq. (6)) is set to 8. We use the $\tau = 8$ nearest SMPL template vertexes’ velocities to get each Gaussian primitive’s local velocity embedding e_i (in Eq. (12)). Additionally, we utilize the adaptive densification in [27] to control the number of Gaussian primitives.

For the DNA-Rendering dataset, we use the SMPL-X [51] model with pose dimension $P \in \mathbb{R}^{25 \times 3}$ in our experiments. We adopt the SMPL-X template with $N = 10475$ vertexes as Gaussian initialization. We optimize 30k iterations and set the loss weights $\lambda_0 = 1.0, \lambda_1 = 0.01, \lambda_2 = 0.01$. The initial sampling step and the increasing interval are $s_0 = 1$ and $\Delta s = 2$, respectively. The final sample scales are $\mathcal{S} = \{1, 3, 5\}$ (in Eq. (14)). For the I3D-Human dataset, we follow Dyco [8] to use the SMPL [43] model in our experiments for fairness. The number of SMPL template vertexes used for Gaussian initialization is $N = 6890$ and the corresponding pose dimension is $P \in \mathbb{R}^{23 \times 3}$. We optimize 15k iterations and set the loss weights $\lambda_0 = 1.0, \lambda_1 = 0.1, \lambda_2 = 0.1$. We set the increas-

ing interval $\Delta s = 9$. The initial sampling step is $s_0 = 24$ and the final sample scales are $\mathcal{S} = \{24, 33, 42\}$ (in Eq. (14)). Note that the experiment in Tab. 4 (d) adopts single sampling step that $\mathcal{S} = \{s_0\}$.

12. Additional Results

12.1. Experiments on ZJU-MoCap

ZJU-MoCap [53] is a common benchmark dataset in human avatar, which is mainly collected under controlled speeds and tight-fitting garments. We use 6 sequences (377, 386, 387, 392, 393, 394) for experiments following the dataset split in 3DGS-Avaar[56].

Comparison on ZJU-MoCap. Results on DNA-Rendering and I3D-Human demonstrate that our methods achieve better performance in complex cases with loose clothing and uncontrolled speeds. Furthermore, Tab. 5 and Fig. 9 show that our method also achieves competitive metrics in simpler controlled scenarios, which verifies the generalizability of the proposed methods.

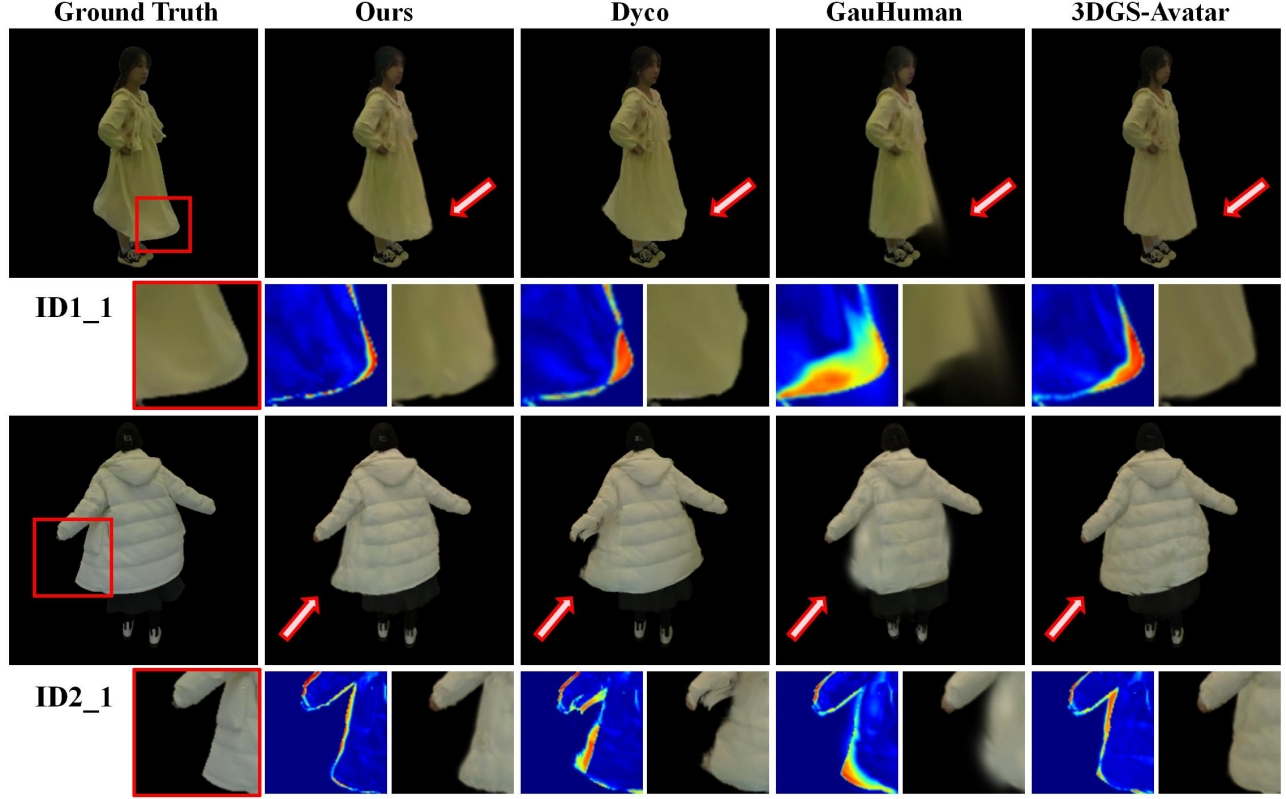


Figure 8. Novel View Qualitative Results on I3D-Human.

Table 5. Quantitative Results on ZJU-MoCap.

Methods	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS* $\downarrow$
Dyco [8]	30.37	0.9599	29.47
3DGS-Avatar [56]	30.59	0.9609	27.09
GauHuman [22]	31.04	0.9620	31.81
Ours	31.02	0.9619	28.89

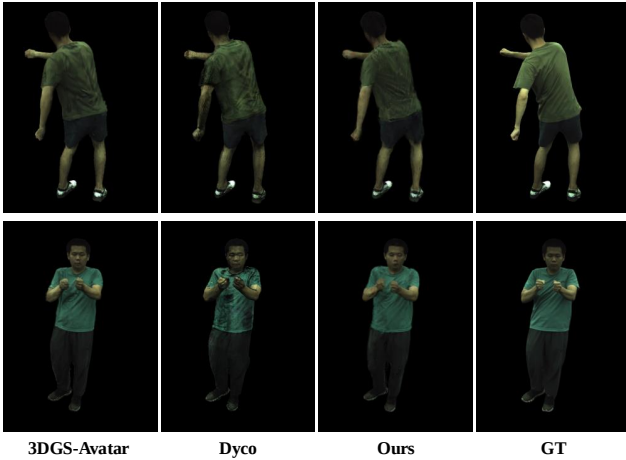


Figure 9. Qualitative Results on ZJU-MoCap.

12.2. Visual Comparisons On DNA-Rendering

Fig. 7 shows additional qualitative comparisons with other SOTA methods on the DNA-Rendering dataset [9]. Our approach excels in regions where motion causes notable appearance variations, producing results that conform to the true shape more accurately, as exemplified by the sleeve (top row) and the skirt hem (bottom row) in Fig. 7.

12.3. Video Quality

We provide rendered videos on the DNA-Rendering dataset [9], along with the corresponding error maps similar to Fig. 3. Please refer to the attached videos for comparison. In the videos, the novel view results are rendered on temporal frames with pose IDs 0–99, using camera view-points that are held out during training.

13. Additional Ablations

Different Sample Steps Tab. 6 shows the quantitative metrics with different sequential sampling steps. The results demonstrate that combining all coarse-to-fine temporal motions as the condition leads to more robust non-rigid deformation and better performance.

Different KNN Sampled Vertexes Tab. 7 presents the quantitative results on I3D-Human dataset [8] with differ-

Table 6. **Impact of Different Sampling Steps on I3D-Human.**

Sampling Step $\mathcal{S}$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS* $\downarrow$
$\mathcal{S} = \{24\}$	32.01	0.9655	30.71
$\mathcal{S} = \{33\}$	32.09	0.9656	30.88
$\mathcal{S} = \{42\}$	32.11	0.9658	30.81
$\mathcal{S} = \{24, 33, 42\}$	32.24	0.9664	29.78

ent numbers of KNN vertexes when deriving the velocity e_i of each Gaussian primitive (Eq. 12). It suggests that considering the motion states of all local regions helps obtain a more robust Gaussian primitive’s velocity embedding, contributing to enhancing performance.

Table 7. **Impact of Different KNN Vertexe Number τ .**

KNN num. τ	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS* $\downarrow$
$\tau = 1$	32.14	0.9659	30.41
$\tau = 5$	32.15	0.9660	30.32
$\tau = 8$	32.24	0.9664	29.79
$\tau = 10$	32.20	0.9663	30.11

14. Computational Cost Analysis

Table 8. **Computational Efficiency.**

Mehods	Train Time	FPS	Train Mem.
Dyco	6 h	0.7	20 GB
3DGS-Avatar	25 m	25	6 GB
Ours ($3k$ iter)	5 m	62	8 GB

We provide computational cost analysis for experiments on ZJU-Mocap (conducted on a single 4090 GPU). Tab. 8 shows that our approach achieves competitive computational efficiency. The additional computation time for each frame’s Vertex Motion Template (multi-scale sampling number $|\mathcal{S}| = 3$, sequence length $L = 8$) is about 0.09 s, and it only needs to be computed once during preprocessing before training.